Product Master Data Governance:
How AI transforms fragmented master data into a single source of truth

A Case Study in Food Manufacturing Company

Diagnosing Product Master Data Challenges from Data governance perspective and Designing an AI-Augmented MDM Solution

A pure MDM centralization initiative hit adoption walls — AI integration emerged as the pragmatic response

Content	Description
Situation	A food manufacturing conglomerate operates an MDM System at only 20% of its potential — 80% of product master data remain decentralized in downstream systems.
Diagnosis	Using DAMA-DMBOK, DQ dimensions, MDM architecture analysis, and Maturity assessment, the root causes were identified: no single source of truth, no formal data stewardship, and a decentralized manual operating model creating redundancy, errors, and delays.
Approach/Obstacles	The organization migrated downstream systems one by one — mirroring each ERP into MDM. This achieved single touch point but inherited ERP constraints, inflated scope with non-master-data fields, and limited standardization to one department's view at a time.
Re-Approach	Shift from system migration to business-driven enrichment — prioritize data topics by stakeholder impact, standardize cross-departmentally using a diagnostic funnel and iterative methodology
Solution	AI-augmented data processing within the cycle — using local LLMs to parse unstructured data (Layer 1) and generate department-specific views (Layer 2), with Data Steward review ensuring governance accountability
Result	Enrichment of Product Master Data fit to serve all business functions while maintaining or reducing data maintenance time, shift from 5-10 to 25-30 centrally governed fields, and a scalable design for enterprise-wide data management

The organization operates a centralized MDM architecture at only 20% of its potential — departments bypass it entirely

Company profile, NPD system overview, and operational assessment

Multiple sectors — many with their own sub-sectors — cross multiple functional domains, making centralized product master data a structural necessity

Company overview:

- One of the **largest agro-industrial and food conglomerates.**
- **Vertically integrated**: Animal Feed, Animal Farm, Food Manufacturing
- 7+ business sectors across the value chain
- 6+ functional domains operating across sectors
- **Case scope: Finished Goods in agro-food business - Poultry, Swine and Processed food sectors**

Scale of complexity:

- **7+ business sectors:** Feed, Aquaculture, Poultry, Swine, Egg, Processed food, and Restaurant.
- **10+ functional domains:** Sales, Supply Chain (SCM), HR, Production, Procurement, Finance & Accounting, and other supporting functions.
- **7+ ERPs and specialized platforms**
- Thousands of active product SKUs
- A single product touches 6+ departments from creation to market

Why this makes MDM critical:

- Each sector has **distinct product specs, regulatory requirements, and supply chains.**
- Each domain has its **own system and data format requirements**
- **Product master data must be consistent** across all active sector
- **Without centralization, each intersection maintains its own version of truth**

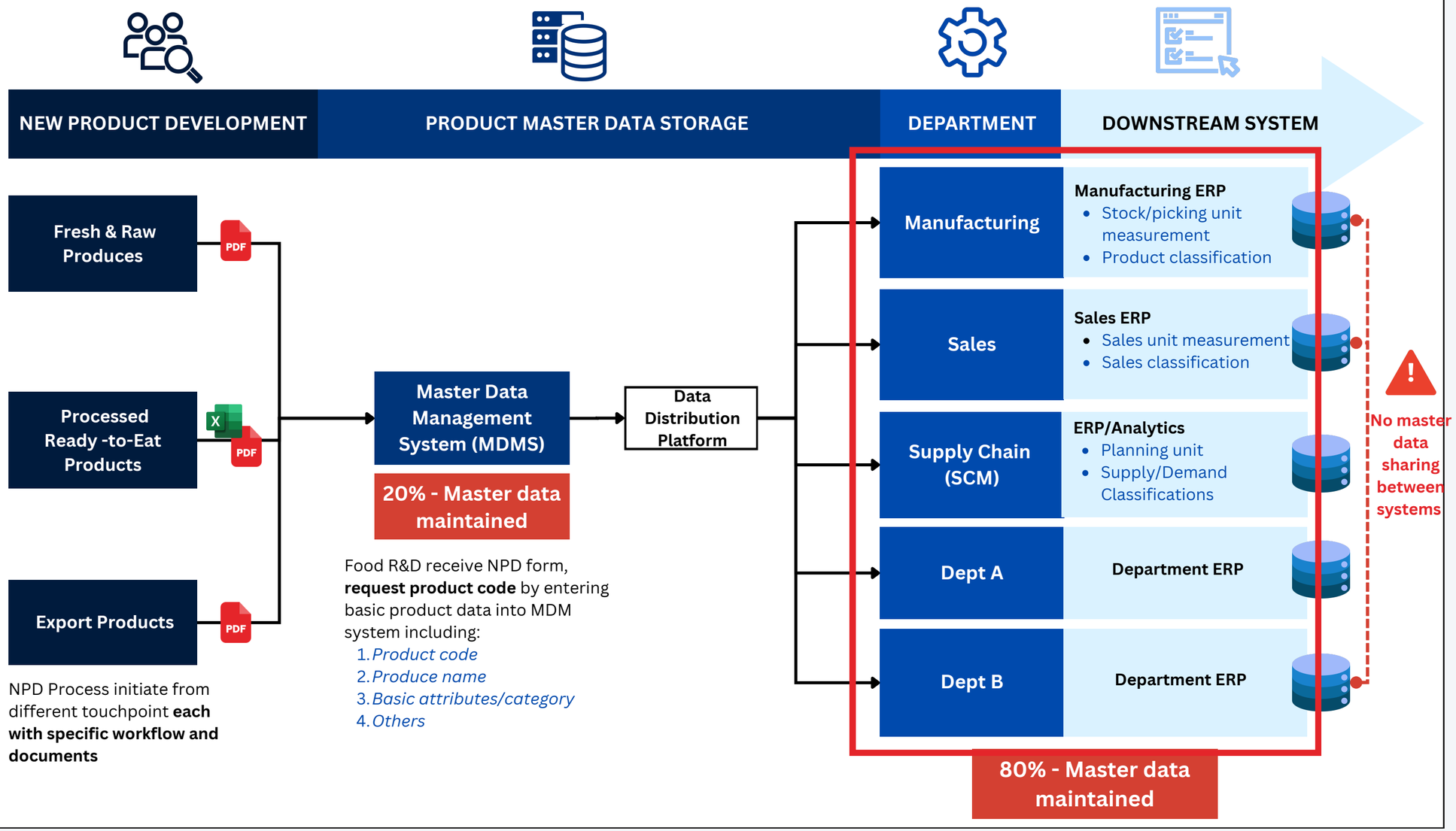
	Functions					
Sectors*	Sales	Production	SCM	Procurement	HR	Finance Accounting
Feed [1)]						
Aquaculture [2)]						
Poultry						
Swine						
Processed Food						
Egg						
Restaurant						

In scope

Not In Scope

- *Each sector-domain intersection requires consistent product master data, the matrix shown is a simplified representation and sub-sectors multiply the actual number of intersections:
  1) Feed has sub sector such as livestock feed.
  2) Aquaculture has sub sector such as shrimp, fish.

Majority of product master data is maintained in downstream systems — MDM stores only product identifiers, not the full profile departments need.



Key Takeaways

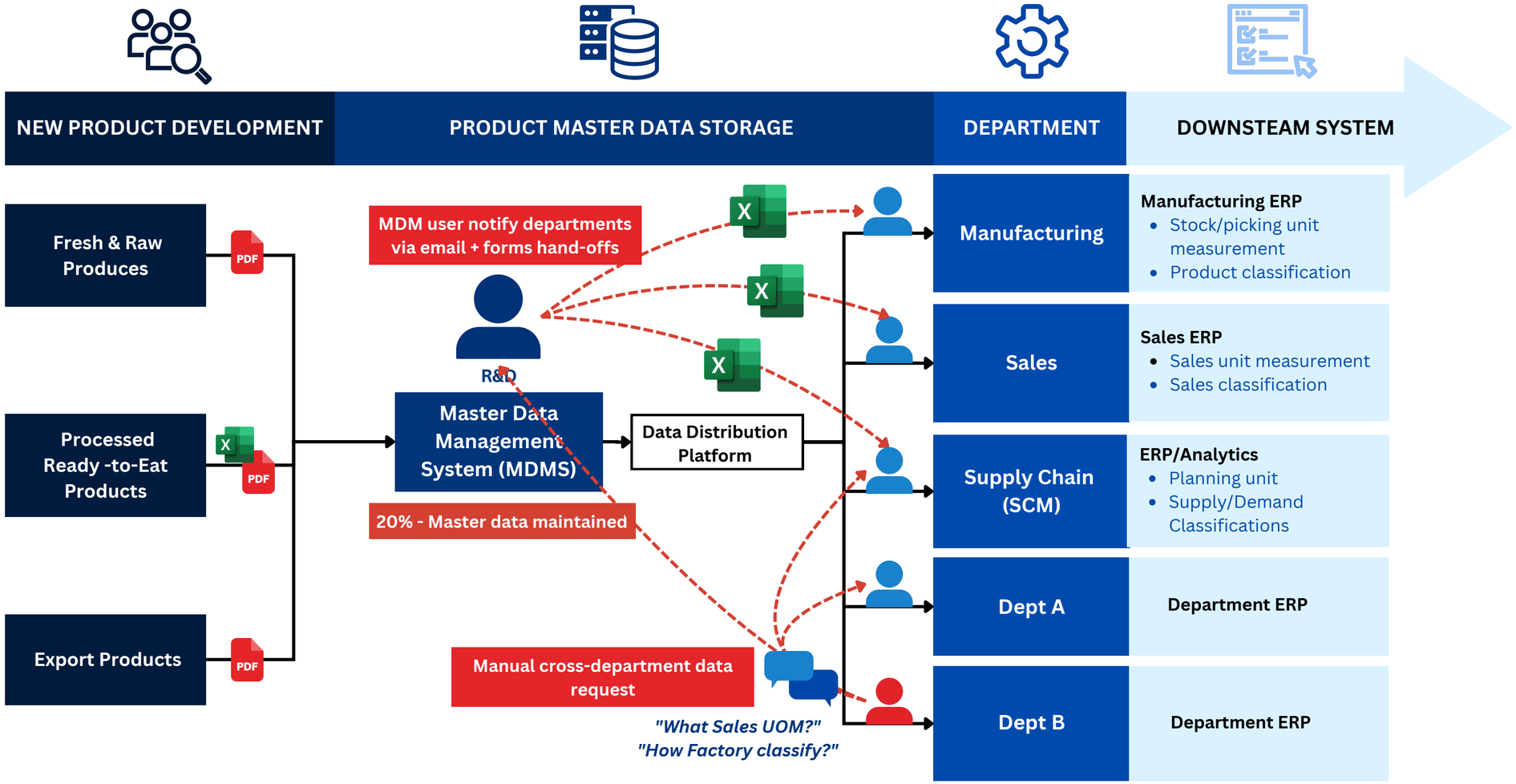
Product master data remains decentralized at downstream systems.

- Department **manually maintains** remaining master data directly into their systems.

Master data is not shared across system, despite all system share similar data - leading to:

1. **No single source of truth**
2. **Redundant data** - entry and updates
3. **Increased risk** of error and misalignment (i.e. spelling, format)
4. **Inefficient operations** causing increased lead time
5. **Lack of transparency -** progress tracking and status.

Manual cross-department coordination, incomplete handoffs, and absence of end-to-end workflow create systemic delays and error risk.



Key Takeaways

Incomplete & Unstructured Data

- R&D enters limited data into MDM, handoffs to departmental users to fill the rest.
- Departments must follow up with R&D via email for detailed specifications, leading to manual transcription errors.

Cross-Department dependencies

- Downstream departments must manually request/receive data from other systems via email/messaging, creating delays.

Fragmented workflow and data incompleteness - leading to:

- No systematic end-to-end workflow
- Manual coordination overhead
- Increased data errors and misalignment risk
- Inefficient operation

A four-lens diagnostic framework reveals six interconnected data governance gaps — tracing to root causes

Using DAMA-DMBOK, DQ dimensions, MDM architecture analysis, and governance maturity assessment

Four diagnostic lenses — from theoretical governance framework to operational maturity — expose where and why the system fails

1 DAMA-DMBOK Framework

The theoretical backbone — governance at center, surrounded by knowledge areas.

Relevant knowledge areas for this case:

1. Document & Content Management
2. Data Model & Design
3. Master & Reference Data
4. Data Quality Management
5. Metadata Management
6. Data Governace

6 knowledge area failing

2 Data Quality Dimensions

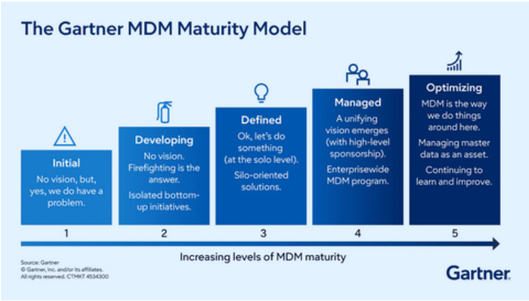
Measuring whether data is fit for purpose across 6 aspects.

Quality distinction:

- **Syntactic quality**
  - tools can check format
- **Semantic quality**
  - only domain experts at source can verify meaning.

Completeness, Consistency, Accuracy risk

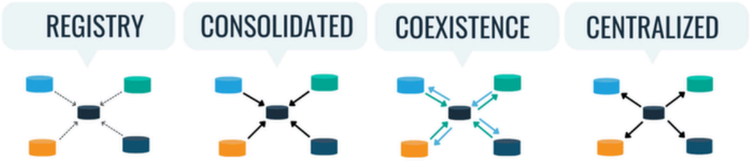
3 Gartner MDM Maturity Model



Strategic stagnancy at Level 2.

4 MDM Architecture Styles

How master data is stored, governed, and distributed across systems.



Technical "Centralization" undermined by "Registry" behavior.

Critical product data is trapped in unstructured documents — and MDM lacks the data model to receive it even if extracted

1.1) Document & Content Management

FINDING

- Critical product data lives **trapped in specification documents (PDFs, Excel)** at the initial stages.
- Documents are upstream data sources that **never get translated into MDM**.
- Discoverable within departments (File storage, Attatchment) but invisible across departments.

ROOT CAUSE

- **No governance policy** requires data from spec documents to be entered into MDM in a format usable by downstream consumers.
- **Requesters aren't required** to enter this data as part of their workflow.

IMPLICATION

To achieve enriched Product Master Data, specification documents must be recognized as governed data sources. Solution needs a way to extract structured data from unstructured documents at scale.

"Product Channel" i need is trapped in documents

1.2) Data Model and Design

FINDING

- MDM data model **only captures product identifiers** (code, name, basic categorizations) not the full product profile.
- **Core product attributes** like storage conditions, packaging configuration, and physical specifications **were never included in the model.**

ROOT CAUSE

- The model was **never entirely assessed** against what downstream departments actually need as source data.
- Aside from product code, name, and few attributes, remainder of existing MDM data model (20%) is a **mirror of one of core department ERP.**

IMPLICATION

- 20% of MDM data model is not enriched enough for sharing
- **MDM must expand from a product registration to a comprehensive product profile** — the foundation that departments requires for their specific operational formats.

but even if freed, MDM has no field to receive it.

Without a complete product profile in MDM, departments create their own data and reference lists — leading to inconsistency, inaccuracy, and uncontrolled definitions

1.3) Master and Reference Data

FINDING

- Since departments **don't rely on MDM as a single source of truth**, each maintains its own product data with their own reference lists.
- The **same concept has different definitions across departments**
  - *e.g., "sales_channel"*
    - **Sales** uses store-level names like Store group A/Store group B
    - **SCM** uses broad categories like Wet Market/Supermarket).

ROOT CAUSE

- With MDM holding only few basic product identifiers, **no department treats it as authoritative.**
- **Without centralized reference data standards**, each department creates their own lookup values — resulting in multiple incompatible formats for the same field.

IMPLICATION

MDM Level: Field-by-field standardization exercise is needed — **consolidating reference lists, agreeing on canonical values**, and building **cross-reference mapping** tables between department-specific formats.

SCM calls it 'Modern Trade.' Sales calls it 'Store Group A.', both are correct but which do i use? (answer: both are correct)

2) Data Quality Management (DQ/DQM)

FINDING

Main **DQ dimensions failing**

- ***Completeness*** - MDM only captures ~20% of product attributes
- ***Consistency*** - Conflicting values for the same product across different systems
- ***Accuracy*** - Data entered by non-domain-specialists, creating accuracy risk.
  - e.g. Finance department maintained Sales and Production views in ERPs.

ROOT CAUSE

- ***Completeness*:** Limited MDM Scope
- ***Consistency:*** Distributed Data Ownership / Master data is updated in silos
- ***Accuracy*:** No policy requiring domain specialists to enter specialist data

IMPLICATION

- **Completeness:** Expand MDM Schema by centralize all decentralized attributes.
- **Consistency**: Lack of automated cross-system synchronization
- **Accuracy:** Assign clear ownership so only subject matter experts can "sign off" on related fields

Who do I ask — R&D, Sales, or Production? And how do I know if what they gave me is correct?

Data lineage and ownership are untraceable — every issue traces back to governance policies that exist on paper but are not operationalized

1.5) Metadata Management

FINDING

- **Knowledge stays tribal** — when attempting to acquire referencing data, people manually track down answers department by department until satisfied.

ROOT CAUSE

- **Field-level lineage for Product Master Data hasn't existed yet**— no way to trace where data comes from, who owns it, or what the business definition is.
- Metadata in dataset level may exist partially in the company datalake, but is **not accessible nor useful.**

IMPLICATION

- **Comprehensive metadata of product master data is essential along with MDM system** — for registering data lineage, field-level ownership, business definitions, and canonical reference values.

Where can I find the real channel data? Is there a map? Who knows best? I talked to Sales — are they really the owner?

1.6) Data Governance

FINDING

- User resistance and low data governance literacy make data governance implementation harder — people don't understand why governance matters.

ROOT CAUSE

- Data governance initiative exists in pilot scope — policies being drafted, not effectivelycommunicated nor operationalized in MDM related business functions.

IMPLICATION

- **Governance policies must be established** to make all master data implications from Cards 1-5 feasible.

Every issue traces back to data governance!

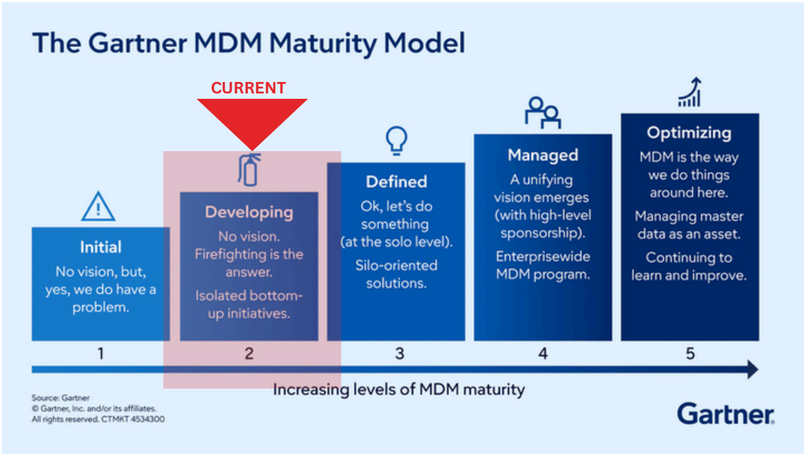
Six interconnected master data management gaps trace back to one root — governance hasn't connected the pieces

Knowledge area	Key findings	Key Implication
Document & Content Management	**Critical data trapped** in spec documents, never enters MDM	Recognize documents as governed data sources; extract at scale
Data Model and Design	MDM is a product registry, **not a full product profile**	Expand model from identifiers to comprehensive product attributes
Master and Reference Data	**Departments maintain their own** data and reference lists	Standardize reference lists, build cross-reference mapping tables
Data Quality	**Incomplete, inconsistent, accuracy** risk from non-specialist entry	Enrich attributes, enforce validation rules, assign role ownership
Metadata management	**limited catalog, no lineage**, no field-level ownership — tribal knowledge	Establish data catalog with lineage, definitions, and ownership
Data Governance	**Initiative exists but not operationalized**	Policies must be established to make all above feasible

No single gap can be fixed in isolation. ***The solution must address all six simultaneously through governance that binds MDM, DQ, and operations together.***

Centralized architecture exists but operates as a registry — governance maturity, not infrastructure, is the constraint.

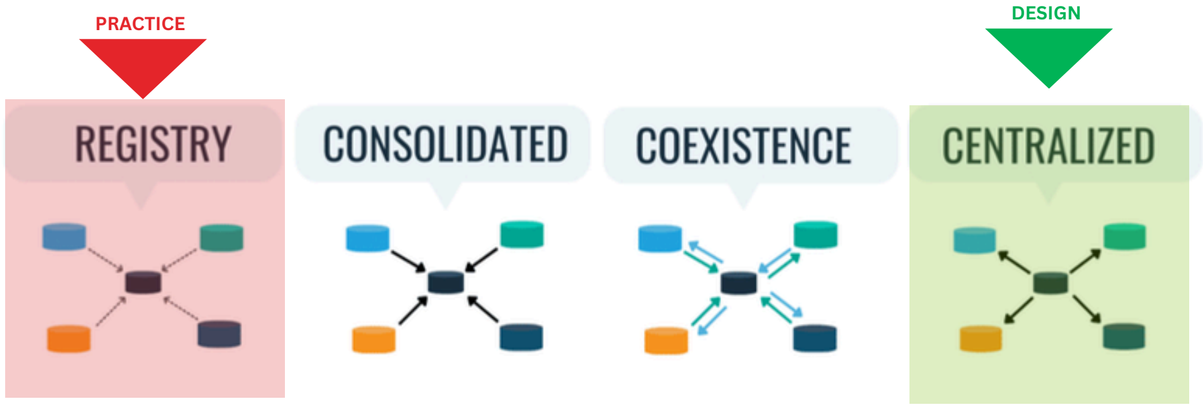
3) Gartner MDM Maturity Model



FINDING

- **Operational Context:** MDM activities reactive and fragmented — triggered by localized issues, not enterprise strategy
- **Functional Gap:** High reliance on shadow systems (Excel/local DBs) departments don't trust the central repository
- **Strategic Stagnation:** Program remains IT-led, failing to provide cross-functional business intelligence.

4) MDM Implementation Styles



FINDING

- **Design vs. Practice:** Infrastructure configured for Centralized — current practice operates as passive Registry
- **Mirror Model:** MDM functions as a "mirror" of a single ERP, inheriting that system's limitations — not an independent golden record
- **Shadow Data:** 80% of product data exists outside the governed pipe — preventing realization of single source of truth

IMPLICATION

- Enrich, don't rebuild: Stop mirroring one ERP. Enrich MDM to be fit-for-purpose — more governed data through the existing pipe.
- Practice must match design. Enriching the MDM data model allows registry practice to gradually move toward centralization — no new infrastructure needed, just more governed data.

MDM, Data Governance, and Data Quality are interdependent — the 'channel' example shows how gaps in one create compliance and operational risk in all three"

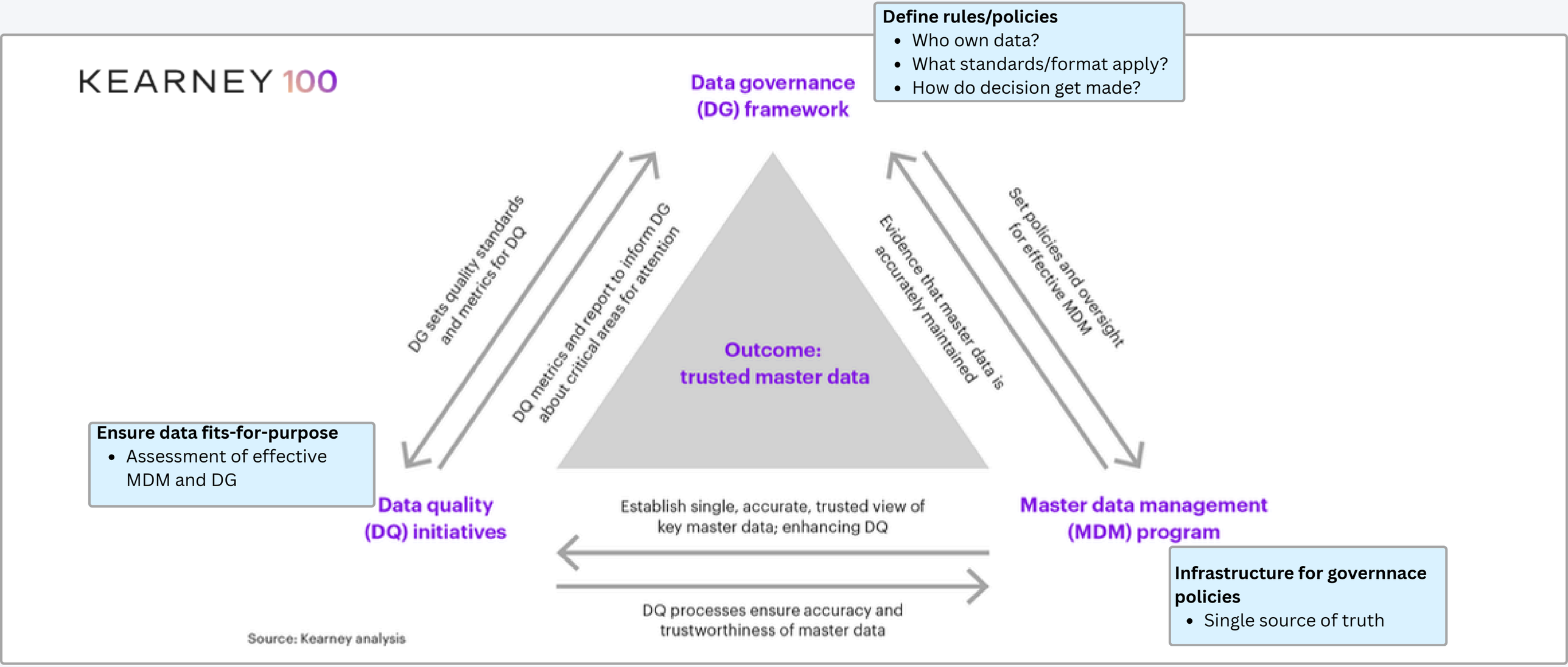


Image source: Kearney Analysis

MDM, Data Governance, and Data Quality are interdependent — the 'channel' example shows how gaps in one create compliance and operational risk in all three"

{”sales_channel” : “modern trade”}

No governance defines the required format per department.
Each department creates their own interpretation:

Department	Context requirement	Data interpretation	Business Risk
Sales	Customer-specific pricing and promotion planning	Modern Trade breaks into • “Brand A” • “Brand B” • “Brand C”	Promotion & Pricing misallocated
Supply Chain (SCM)	Demand planning and distribution routing by channel category	"Modern Trade" (Category Level)	Forecast inaccuracy over/under stock at channel level
Finance	Revenue reporting and by channel	"Modern Trade" by Region	Regional performance misreported based on incorrect geography

MDM - DG - DQ Aligned

Data Governance Policy

- Product Data Owner is **accountable for completeness**
- MDM stores the standard channel category (MODERN TRADE) and **requires governed supplemental fields:**
  - ***channel*** *+ (brand + region)*

MDM enforces:

- MDM System **supports entries of supplemental data.**
- Each department **receives the governed info** from the product spec via MDM — not their own interpretation:

SCM: "Modern Trade"
Sales: "Modern Trade” + “Brand A"
Finance: "Modern Trade” + “Region A"

Data Quality Assessment:

- **Completeness:** Are 3 MDM-fields populated?
- **Consistency:** Are Department fields logically match MDM category. *(e.g. Brand A is always Modern Trade)*
- **Accuracy:** Did a specialist input this?

Master data governance gaps cascade into four impact tiers — from consumer safety exposure to financial waste from structural workarounds

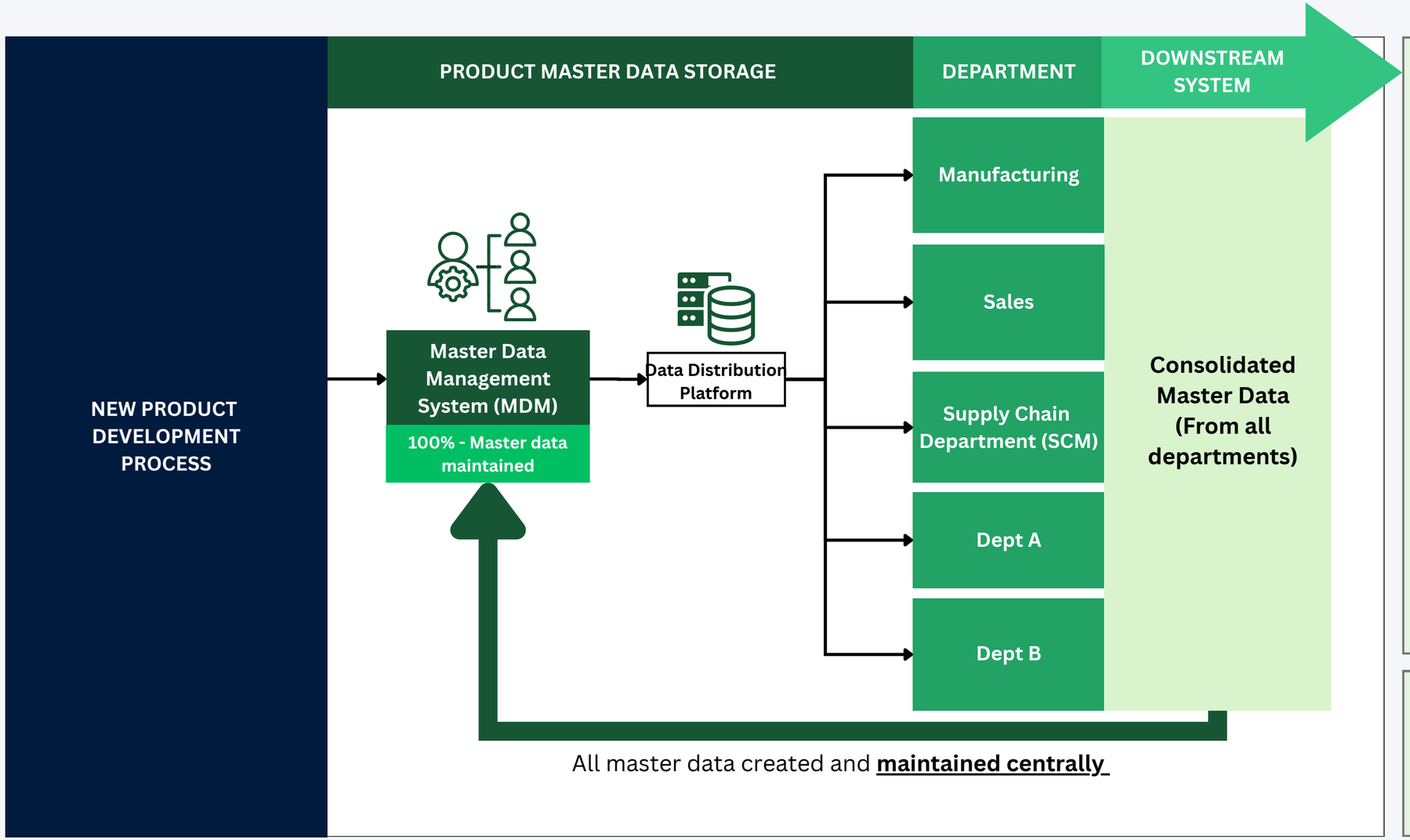
1	**RISK** Manual processes without systematic validation of critical fields	**CONSEQUENCE** • **Consumer safety** exposure — i.e. allergen requirements at risk of mislabeling • **Regulation safety** exposure — i.e. halal certificates, export-Controlled compliance
2	**OPERATIONAL** Incomplete MDM forces manual data acquisition across departments	**CONSEQUENCE** • **Extended NPD cycle means lost Sales, missed market window** ◦ i.e. Analytics team can't make demand forecast on time because they must manually chase product attributes across departments (~1-2+ days per product)
3	**STRATEGIC** Shadow data creates inconsistent views across departments	**CONSEQUENCE** • Product portfolio, market direction, and financial planning & analytics **built on unreliable consolidated data, risking misdirected company strategy**
4	**FINANCIAL** Ungoverned MDM forces workarounds and reactive correction	**CONSEQUENCE** • **Unnecessary Effort**: Workers chase missing attributes across teams • **Error Correction Cost:** Staff reconcile conflicting records when discrepancies surface • **Excessive Hiring:** Additional staff hires to cope with volume

Product master data integrity is business critical — gaps open for human error, cascading to adverse impact at enterprise level

The organization's response: migrate downstream systems into MDM

Vision - MDM as a Single Source of Truth

The strategic objective - migrate all downstream product data to MDM, standardize fields, and finally, block creation in downstream systems.



1. **Data Centralization** - **Migrate all downstream ERP product set-up** to MDM
   - **Standardize fields** across departments
   - Downstream systems only consume master data from MDM
   - Block create/modify capabilities in downstream systems

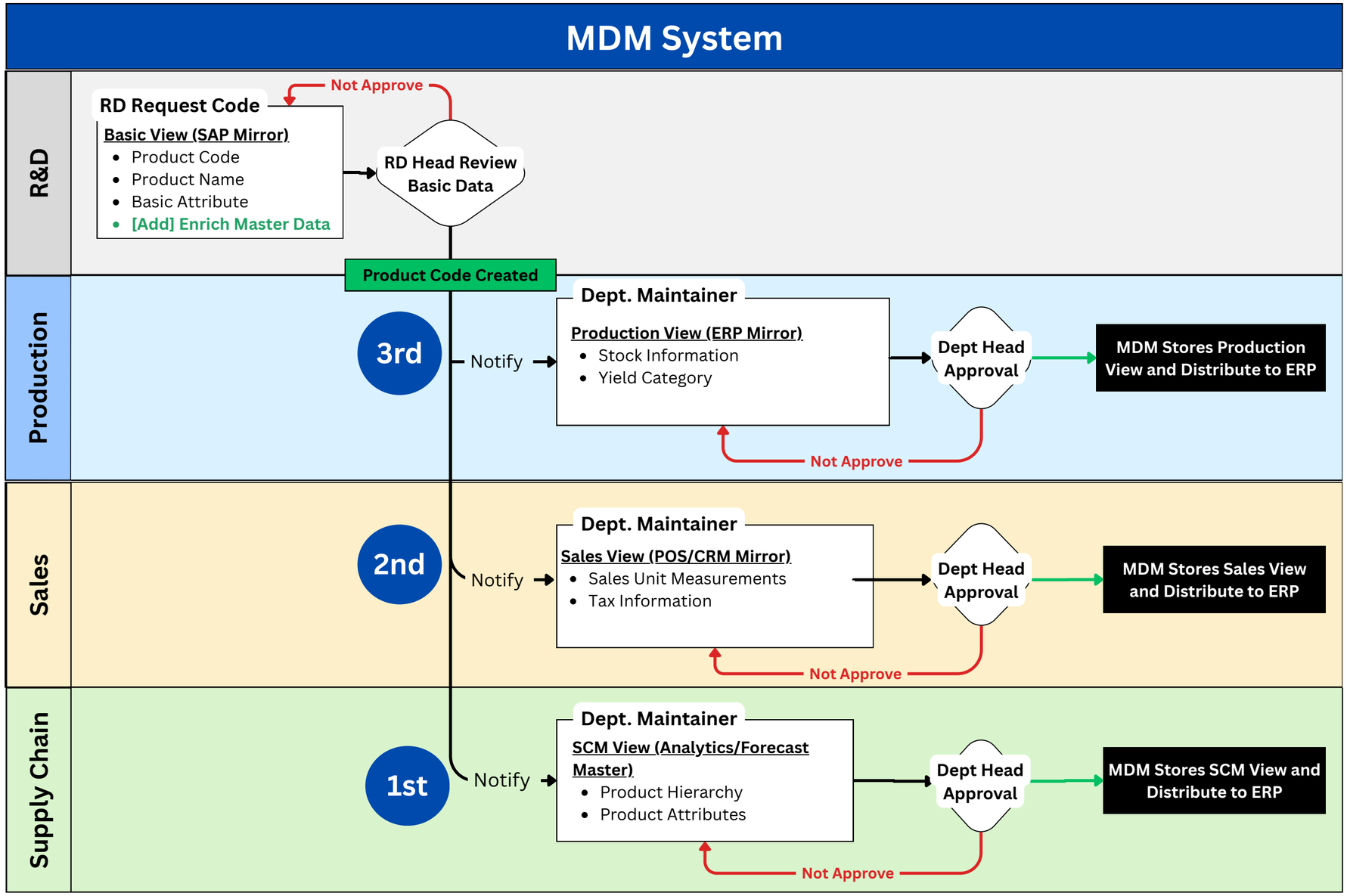
2. **MDM Process Standardization** - Design end-to-end process from creation to deletion in MDM
   - Define clear roles.
   - Optimize data entry sequence
   - Data validation rules
   - Ensure data quality at source

Expected Result

- Single source of truth
- Standardizes MDM process

Note: This was a purely organizational/process initiative — no AI components were part of the original design.

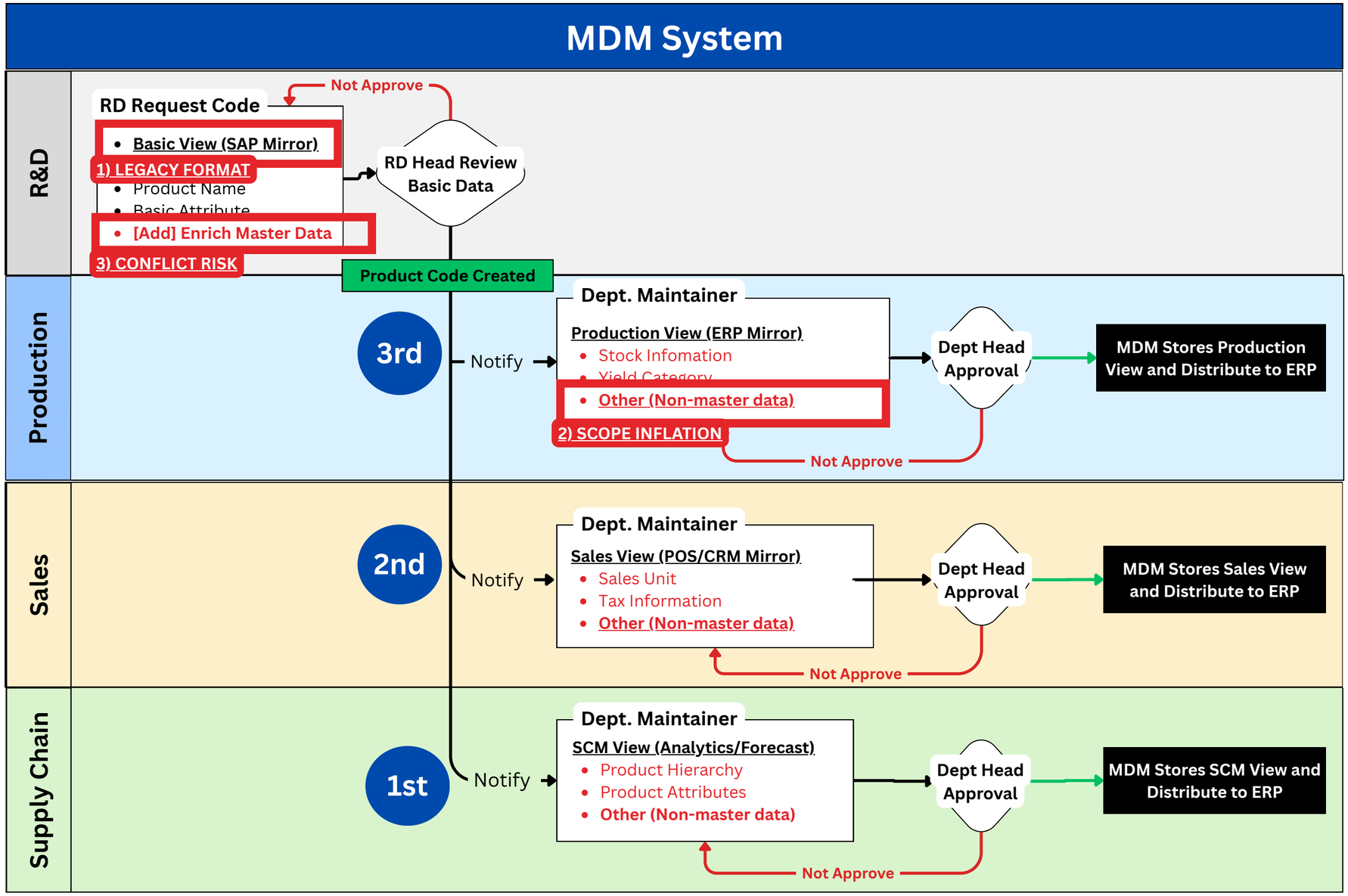
The organization migrated downstream systems one by one — each view mirroring its source ERP to enable single-entry for departments



ORGANIZATION INTENDED METHOD:

- **Migration by department (system)**
  - Understand how they enter data into their ERP to operate.
- **Get full field list:** datatype, mandatory/optional, reference tables, and etc.
- **Assess which fields qualify as master data** based on Data Governance team and department input
  - e.g. unit measurement should be master data, It should be input it via MDM instead of their own system
- **Standardize identified master data**
  - Redesign data model, enrich at MDM level (R&D as entry point)
- **Mirror remaining operational fields as-is**
  - Department inputs in one system only (MDM acts as mirror of their ERP)

System-by-system migration inherits ERP constraints into MDM — limiting standardization and inflating scope with non-master-data fields



OBSTACLES

1) Legacy format is locked

- Basic View mirrors SAP ERP format
- R&D depends on these fields for their SAP workflow.
- Changing the data model breaks the MDM-to-ERP interface.
- **Standardization must work around it, not replace it.**

2) Scope inflation with non-master-data fields

- To achieve single-entry, MDM must carry all fields a department's systems expects. -
  - Operational fields and Reference data that aren't truly master data.
- MDM **becomes a container for ERP fields, not a governed master data repository.**
- Harder to identify what actually needs standardization.

3) Standardization biased

- Standardization effort **focuses on that department's needs** — not cross-departmental requirements
- Data model **designed from limited departmental inputs**
- **Cannot see the full picture** of how a data topic (e.g., master unit measurement) flows across all departments.

Shift from system migration to business-driven enrichment — prioritize data topics by stakeholder impact, not ERP sequence

What the obstacles taught us

Centralization

Not all migrated fields are master data

- System-by-system migration forces **MDM to carry operational fields** that **add no cross-departmental value.**
- Enrichment should **focus on fields that serve multiple departments.**

Standardization

Standardization must be cross-departmental

- Standardization only succeeded when **treated as its own topic** — touching all relevant departments simultaneously.
- **Scoping by field cluster** makes dependencies visible and scope manageable.

Communiucation

R&D work increases regardless, but system migration obscures why

- 80% of enriched fields naturally trace back to R&D as subject matter expert (SME)
- System-by-system migration buries this reality under scope inflation — R&D sees a growing list of fields without understanding which ones matter and why.

The revised approach

Trace from business outcome to data requirement.

- **Start with what stakeholders need** — Sales needs revenue, SCM needs planning accuracy, compliance needs safety.
- **Work backward** to which data fields must be accurate, complete, and governed in MDM.

Enrich topic by topic, not system by system

- **Each cycle addresses one data topic** (unit measurement, storage temperature, product hierarchy) across all affected departmental views.
- **Scope is manageable, impact is measurable, dependencies are visible.**

Align MDM with business outcomes, not governance compliance.

- **Connect each data topic to the business result it enables**.
- R&D understands purpose when fields link to Sales fulfillment and SCM planning.

MDM as ongoing discipline, not one-time project.

- Each business need generates the next enrichment cycle.
  - New export market → new regulatory fields → next cycle.
- **MDM evolves with the business.**

A three-tier model prioritizes master data topics by business consequence — from safety-critical through operational to value-added

Tier	Category	Definition	Case's Example	"Fit-for-Purpose" Response
0	**Safety/Regulatory**	*Data that if wrong causes harm or legal violation*	• Allergen declarations • Halal certification • Regulatory Data	Cannot proceed — must fix before data moves forward
1	**Operational**	*Data that if wrong or missing, blocks systems and transactions*	• Product weight for POS • Unit of measurement • Tax classification	Business operations partially blocked — fix urgently
2	**Value-added**	*Data that if wrong degrades analytics and decision-making*	• Analytics team wanting granular storage temperature categories (chill to chill+ambient) • Product classification for reporting	Business continues with workarounds — improve gradually

Fit for Purpose:

- **Data quality is not measured by perfection**, but by whether it meets the threshold its business use demands.
- Master Data Quality is not about the field itself — it's about the **consequence of getting it wrong for each specific use**

Two discovery streams — reactive complaints and proactive profiling — scope which data topic to standardize next and why

REACTIVE INPUT	PROACTIVE INPUT
Stakeholder report pain points	1. Data Profiling 2. Stakeholder Engagement

STEP 1: PRELIMINARY ASSESSMENT
"What happened, and how big is it?"

Task: Log the issue, gather context
- what, when, where, who, how big

Output: issue log with scoped boundary.

STEP 2: DEEP & BROAD INVESTIGATION
"Trace the problem beyond the reporter - who else does this touch"

Task: Drill into as-is processes and systems, trace how this data topic flows across all departments, apply DQ dimensions.
Output: Full diagnostic (Root cause, affected parties, implication)

STEP 3: CLASSIFICATION
"How urgent is this?"

Task: Assess urgency using three-tier model — Tier 0 (safety) / Tier 1 (operational) / Tier 2 (value-added)
Output: Prioritized issue register with tier per issue

STEP 4: VALIDATION
"Do affected departments agree with what we found?"

Task: Present findings to all affected departments, confirm scope, refine classifications, surface adjacent issues.
Output: Stakeholder-validated, aligned assessment

STEP 5: ACTION PLAN
"What do we do about it, and who decides?"

Task: Package into decision-ready recommendation with prioritized sequence
Output:
- **If Agreed** → Top-priority topic proceeds, remaining topics queued for next cycle
- **If Inconclusive** → Escalate to leadership

Output feeds A→B→C execution cycle (next slide)

Each A→B→C cycle standardizes one data topic across all affected views — scoped by the diagnostic funnel (How to execute it)

Scoped and prioritized topic from diagnostic funnel

A — Discovery, diagnosis, and prioritization

- Receive prioritized data topic from diagnostic funnel (e.g., unit measurement)
- Map how this topic currently works across all departmental views — not just the reporting department
- Identify cross-view format dependencies
- Assess lifecycle complexity (e.g., modifying units on active products, reference data cleansing)
- Synthesize problem and root cause, validate with stakeholders

B — Data consolidation, MDM standardization, and workflow adjustments

- Consolidate how the topic is used across views — common fields, format mismatches, mapping requirements
- Design canonical data model — field definitions, constraints, cross-system mapping
- Address reference data dependencies (cleansing, consolidation)
- Where field complexity requires it
  - AI-assisted generation for format translation and standardization
- Define metadata and field-level lineage
- Adjust workflow dependency chain if necessary

C — Implementation, launch, and improvement

- Implement solution — new data model, updated MDM UI, validation rules
- UAT testing across all impacted departmental workflows — including departments that didn't report the original issue
- Train-the-trainer with key representatives from each affected department
- Pilot deployment — measure adoption and time-to-data-readiness
- Result: One more data topic governed centrally with cross-departmental standard

Next cycle (New Topic)

Next Cycle : storage temperature → product hierarchy....

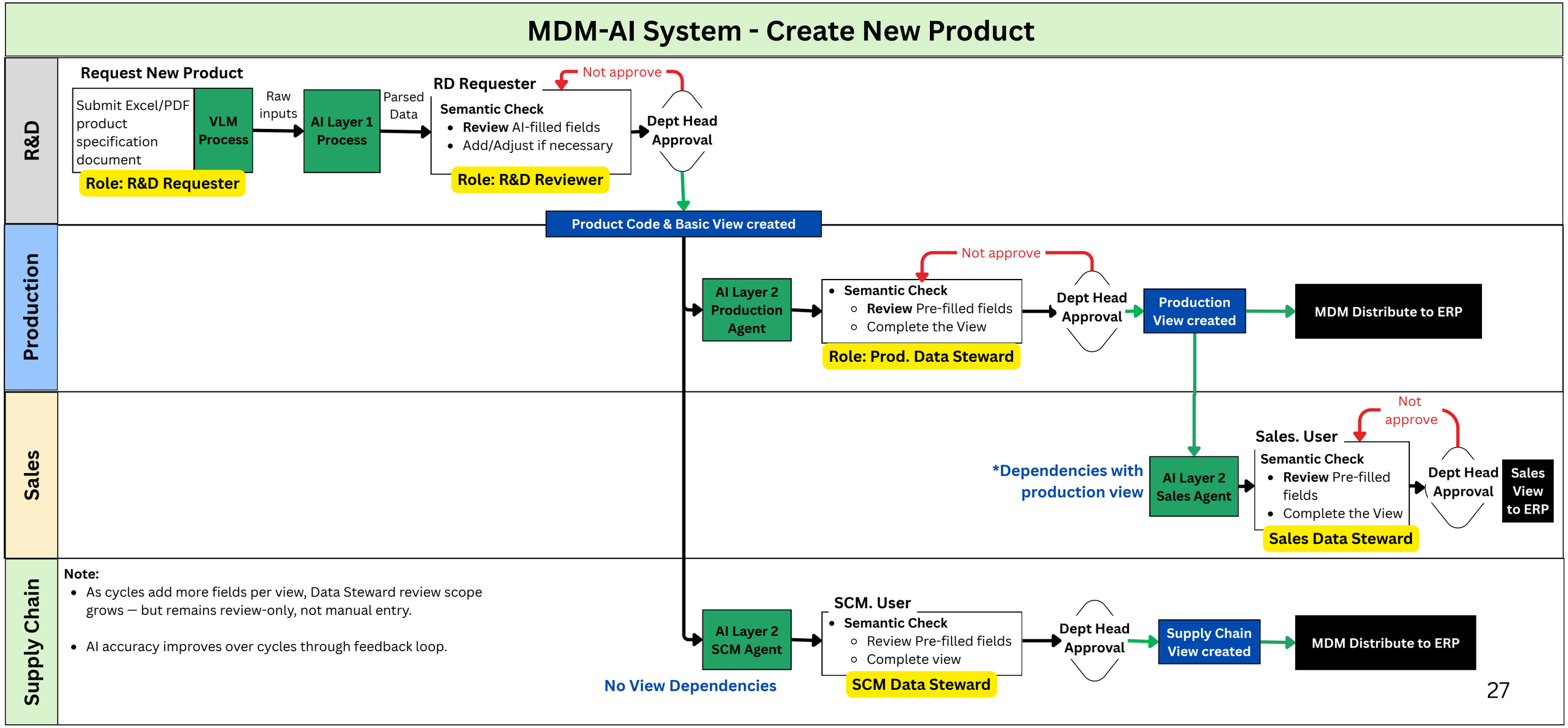
Each cycle makes MDM more valuable to more departments — **adoption grows from demonstrated value, not mandated compliance**.

Personal Project

Developed independently to systematically approach product master data centralization — created from the challenges encountered.

AI-MDM Integrated — A proof-of-concept field-by-field standardization

AI-MDM workflow: AI generates, Data Stewards validate, MDM distributes — achieving centralization without increasing manual effort



Enrichment obstacle: R&D workload triples from 5-10 to 25-30 fields — AI auto-fill reduces actual effort to review-only

Discovery

Responsibilities shifts

- Many additional MDM fields trace back to R&D responsibilities (i.e., unit measurement)
- R&D must maintain additional data on MDM
- R&D workload increases (5-10 fields → 25-30 fields)

Impact

Adoption difficulties

- Though these SHOULD be R&D's responsibility as subject matter expert, **perception of more workload increase threatens adoption**

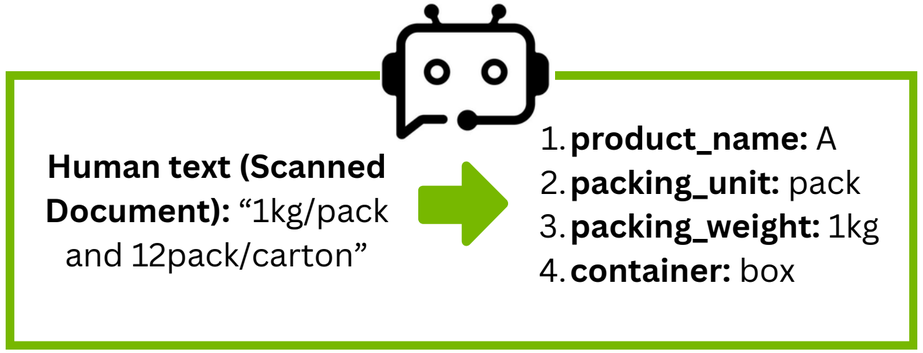
AI Solution

Layer 1: AI auto-fills comprehensive master data from minimal inputs

- AI parses unstructured text → structured fields
- Pre-fills 25-30 fields automatically
- R&D remains accountable for data quality — reviews AI suggestions before approval
- Human-in-the-loop ensures reliability

Result:

- MDM enrichment achieved without high workload increase
- Entry time reduction (15-20 min review vs 30-45 min manual)
- R&D effort: review-only, not 100% manual data entry



Standardization obstacle: MDM generic data is insufficient for specialized departments — AI translates to department-specific formats automatically.

Discovery

Existing MDM master data are not universal.

- MDM stores data at generic level
- Different departments **require different levels of detail:**
  - **MDM** contains "CHILL/FROZEN"
  - **SCM** requires "FROZEN" and "-18°C ± 2°C" (insufficient from MDM)
  - **Sales** requires "AMBIENT/CHILL/FROZEN" (insufficient from MDM)

Impact

Consolidation challenges

- **Fit-for-purpose:** how to design MDM product master data that serve all?
  - **Too generic** = Not useful for specialized departments (SCM, Finance)
  - **Too detailed** = Overwhelming for R&D to maintain
  - Multiple format variations unmanageable in single system
- If not addressed, even if Master Data exist for referencing, departments will revert to ad-hoc siloed master data

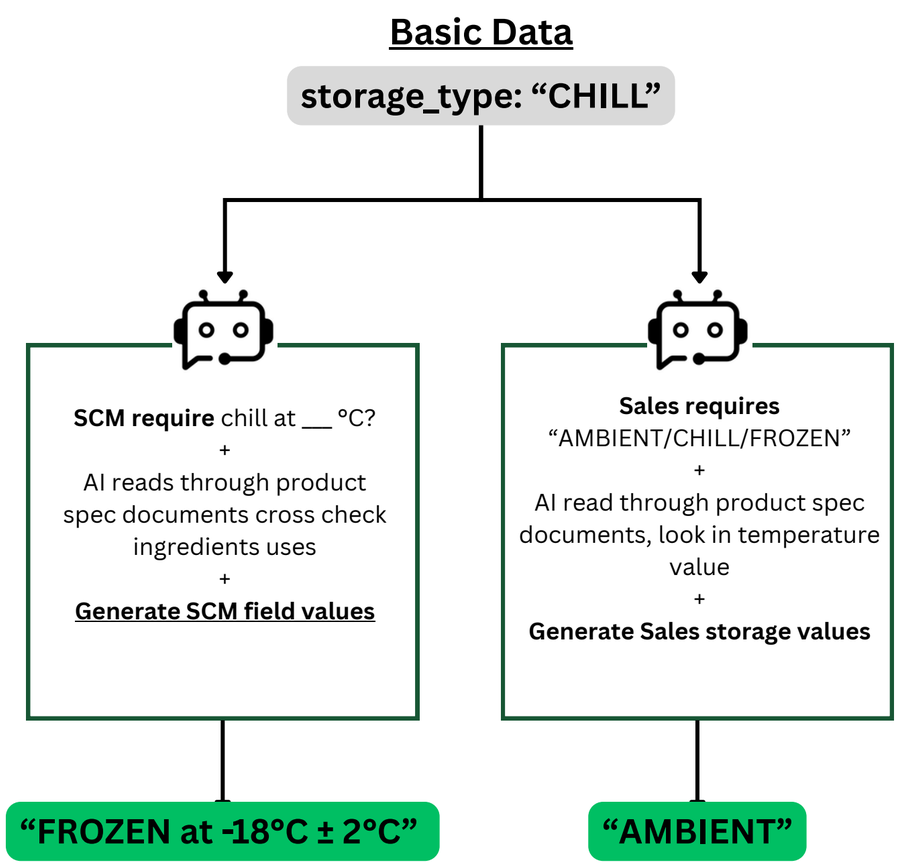
AI Solution

Layer 2: AI translates MDM generic data to department specific formats

- MDM stores generic master data (manageable for R&D)
- AI understands each department's granularity requirements
- Automatically translates to specialized formats

Result:

- R&D maintains simple, generic data
- Departments receive data in their required format
- Minimizes human effort to interpret master data
- Keeps variations synchronized with MDM source



Within AI Layer 2: MDM field to department context transformation is assessed based on complexity into 3 tiers

Tier	Transformation Method	How it works	Technique
A - Low Complexity	Deterministic Mapping:	**Direct value lookup, same input always produces same output.** • No AI used	• YAML config lookup tables. • Direct key-value mapping • Python dictionary lookup • no logic, no AI.
B - Medium Complexity	Rule-Based Transformation:	**Deterministic but with business rules.** • Conditional logic, unit conversions, format transformations defined in config. • No AI used	• Python rule engine with YAML config. • Conditional logic
C - High Complexity	LLM-Assisted Contextual Generation:	**For fields where the output requires interpreting data through the department's specific context.** • The LLM is loaded with that department's context from config file to generate appropriate values. • These fields are flagged for human review.	• Local LLM (Ollama) with RAG (if necessary) • Department context loaded from YAML config as prompt context • LLM interprets product data against department requirements to generate values

Deterministic-first, LLM-second

AI is only used where rule-based logic cannot determine the output. This minimizes unpredictability and maximizes auditability

AI augmentation selected as the only approach that reduces effort without compromising governance accountability.

NOT VIABLE SIMPLIFYING MASTER DATA MODEL	VS	VIABLE: AI AUGMENTATION
Undermines the strategic objective • Reduce 25-30 fields back to manageable level • Accept less comprehensive product profile. **Treats symptoms, not root cause** • Might not solve granularity mismatch if too general		• AI absorbs the effort increase • Domain experts remain accountable (review, not entry) • Governance integrity preserved (human approval gates at every stage) • Configurable and scalable (iterative process)
Reduces entry workload but defeats centralization purpose. **Departments revert to shadow data for missing fields. DQ issues persist.**		**Preserves governance accountability while making centralization operationally viable**

Success metrics aligned to business impact — measuring whether the solution addresses what matters.

Business Impact	What we said was at risk	Success Metric	How to measure
Risk	Consumer safety exposure from incorrect product attributes	**Tier 0 field accuracy rate**	• **Percentage** of safety-critical fields (allergens, storage, halal) correct at first pass
Operational	Extended NPD cycle, lost sales, missed market window	**Data readiness time**	• **Time taken** from product code creation to all views approved. • **Number of products completing full workflow** per period vs before
Strategic	Decisions built on unreliable consolidated data	**Product-level first-pass acceptance rate**	• Percentage of products where **all views approved without modification** across entire dependency chain
Financial	Labor spent on redundant maintenance and manual re-keying and resolving error occur	**Manual effort reduction**	• **Minutes per product**: manual entry (before) vs review-only (after). • **Rework requests after approval** — how often downstream users report issues post-activation

The AI-augmented solution handles data generation and human-in-the-loop review — limitations remain for further exploration

Recap: What the project covers

CENTRALIZATION APPROACH

Iterative improvement

- Each cycle expands standardized field coverage

AI-MDM POC

- **AI-assisted basic data** generation from NPD documents (Layer 1)
- **AI-assisted department view** generation with dependency chain (Layer 2)
- **Data Quality - Accuracy**
  - Human-in-the-loop review ensuring domain expert accountability
  - Cascading approved data through sequential workflow

Known Limitations/Scope Exclusions

SCOPE CONSTRAINTS

- **POC, not production**:
  - POC, not production — validates that AI can generate reviewable output. Review UI is a systems implementation concern
  - Components validated individually. Full end-to-end integration testing not implemented

- **ERP interface is out of scope:** The POC stops at generating departmental views. Actual integration to downstream ERPs via middleware (e.g., Kafka) is outside the project boundary.

- **Modifying Existing Product Master Data is out of scope:** It is depending on nature of the organization, but in general, given that we have made connection of master data across organization systems via MDM, modifying active record usually require cross-functional assessment.

SYNTHETIC REFERENCE DATA

- All reference data reconstructed from domain experience, not extracted from actual production systems

AI ACCURACY

- Would improve significantly with access to real historical product databases
- Results are feasibility indicators, not production-accuracy benchmarks

Supporting Slides

Technical Detail: VLM extracts structured data from scanned NPD documents using coordinate-based spatial mapping

VLM Process

INPUT: DOCUMENT

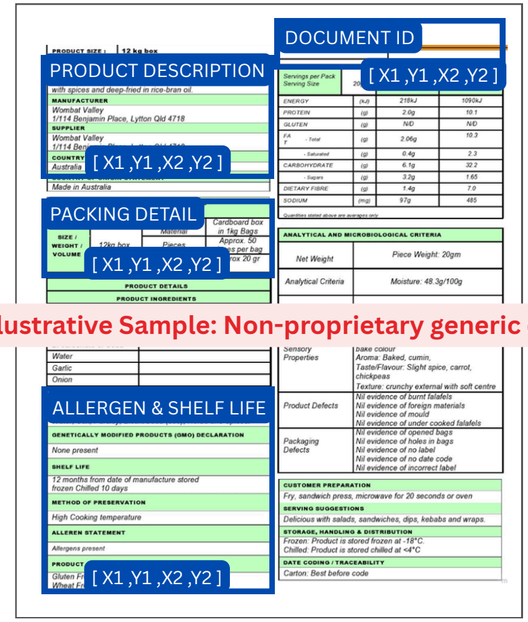
1. **Coordinate Anchoring:** Predefined coordinate mapping for key data fields.
2. **Noise Reduction:** Isolates target pixels to eliminate background interference.
3. **Layout Blueprint:** Creates a reusable spatial map for standardized form structures.

TARGETED VLM EXTRACTION

1. **Snippet Ingestion:** Feeds localized image crops directly to the Local VLM (Qwen3).
2. **Prompt-Driven Logic:** Pairs each crop with specific instructions.
3. **Contextual Extraction:** Simultaneously identifies multiple attributes (Net Weight, Packing, etfrom a single image crop.

GET STRUCTURED DATA OBJECT

- Converts raw VLM extracts into a **structured object for immediate programmatic handling.**



*Illustrative Sample: Non-proprietary generic data.

Local VLM, Qwen3

Input: *coodinate [70, 251, 2839, 1030]*

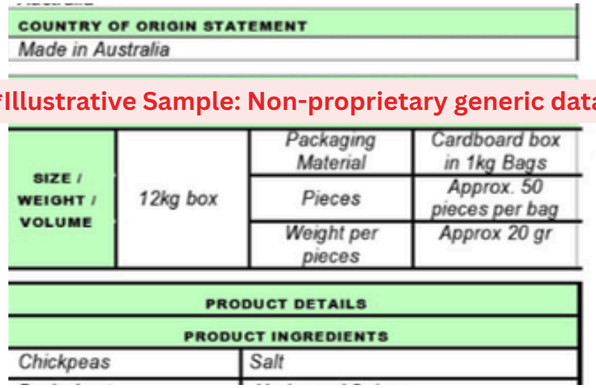
Prompt:

"Extract the following information:

1. Product Name: <string>

2. Spec/Sample No: <string>

3. Packing Detail: <string>"



COUNTRY OF ORIGIN STATEMENT

Made in Australia

*Illustrative Sample: Non-proprietary generic data.

SIZE / WEIGHT / VOLUME	12kg box	Packaging Material	Cardboard box in 1kg Bags
		Pieces	Approx. 50 pieces per bag
		Weight per pieces	Approx 20 gr

PRODUCT DETAILS	
PRODUCT INGREDIENTS	
Chickpeas	Salt

```
result = {
    'product_name': 'ABC',
    'spec_no': 'AA-123',
    'packing_detail': '3kgs/bagx8bags/carton'
}
```

Technical Detail: AI Layer 1 transforms packaging free text into structured unit master data

AI Layer 1
Stage 1 - Parsing Unstructured Text

CASE EXAMPLE INPUT

Input: Packing details as free text

"400g/pack and 12 pack/carton"

LOAD RESOURCES

- CSV file of **curated Packing detail samples** with parsed result ***(Preview)***
- **Unit Mapping:** Translate Human Unit to System Unit ***(Preview)***

AI PROCESSING Ollama

Method Use: RAG

- Parses **unstructured text to structured fields**
- Pre-fills MDM fields automatically

OUTPUT

Structured MDM Basic Data

- 'base_unit' = 'KG'
- 'inner_unit' = 'PAC'
- 'inner_weight' = 0.4
- 'outer_unit' = 'CAR'
- 'inner_per_outer' = 12
- 'outer_weight' = 4.6

Technical Detail: AI Layer 1 classifies product hierarchy using semantic similarity matching

AI Layer 1 Stage - 2 Product Hierarchy Classification

Structure
pg1: Grade
pg2: Parts
pg3: Product Use
pg4: Packing
pg5: Status
pg6: Categoty

LOAD RESOURCES

1. Curated MDM-stored Product Master Data ***(Preview)***
2. Level 1-6 Reference Data ***(Preview)***

PROCESSING

1. **Semantic RAG Matching**: **Product name embedding similarity search** against the Curated Product Master Data samples.
2. **Hierarchical Alignment**: Retrieve the top-ranked match score (0-1)
3. **Similarity threshold condition:**

IF High Similarity(>=0.80) THEN

Ollama

1. **Inherit:** NPD Inherit top scorer pattern.
2. **Contextual Refinement:** AI Read the new product's basic data and adjust attributes to align with the retrieved master data standards.
3. **Confidence Scoring** (0-1), in each level for review

IF Low Similarity(<0.80): THEN

1. **No AI used**
2. **Flag NPD as a "New Entity"** for R&D to manually enter data and Review to ensure data integrity.

OUTPUT

High Similarity

- pg1 = **WHOLE S** (confidence 0.95)
- pg2 = **WHOLE** (confidence 0.95)
- pg3 = **MAIN PRODUCT** (confidence 0.95)
- pg4 = **BAG** (confidence 0.95)
- pg5 = **FROZEN** (confidence 0.95)
- pg6 = **CHICKEN SLAUGHTER** (confidence 0.95)

Low Similarity

- pg1 = NONE
- pg2 = NONE
- pg3 = NONE
- pg4 = NONE
- pg5 = NONE
- pg6 = NONE

Technical Detail: Layer 2 generates Sales view using approved basic data and production view as inputs

Generating Sales View

INPUT

1. Basic View (from Layer 1)
2. Production View (prerequisite view)

Basic View
material_desc:
"Grilled Chicken Breast Teriyaki Flavor 200g"

LOAD RESOURCES

- Departmental Config (External Context for AI to incorporate into reasonimg)
  - POS Shortname abbreviation **(Preview)**
  - Unit Format Conversion **(Preview)**

AI PROCESSING

- Agent takes approved basic data from Layer 1 and **generates department-specific views.**
- Each view is **tailored to what that department's system needs**
  - Views with dependencies must wait for prerequisite views approval before initiating
- Generated View are **undergo approval process** before stored and distributed to ERPs

OUTPUT

pos_name_en:
"Grl Chkn Brst Teri"
pos_name_th:
"อกไก่ย.เทอริยากิ"